LAB ASSIGNMENT 7

PROBLEM 1: Apply the GAUSSIAN NAIVE BAYES CLASSIFIER on all three given datasets and analyze their performance.

CODE:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

df_d = pd.read_csv('diabetes_dataset.csv')
print("Diabetes")
print(df_d.head())
print("Missing values")
print(df_d.isnull().sum())
df_d.fillna(df_d.mean(numeric_only=True), inplace=True)
df_s = pd.read_csv('Social_Network_Ads.csv')
print("\nSocial Network Ads")
print(df_s.head())
print("Missing values")
print(df_s.isnull().sum())
df_s.fillna(df_s.mean(numeric_only=True), inplace=True)
df_t = pd.read_csv('titanic.csv')
print("\nTitanic")
print(df_t.head())
print("Missing values")
print(df_t.isnull().sum())
df_t['Age'] = df_t['Age'].fillna(df_t['Age'].median())
df_t['Embarked'] =
df_t['Embarked'].fillna(df_t['Embarked'].mode()[0])
df_t = df_t.drop(columns=['Cabin'])
print("Missing values (after analysis)")
print(df_t.isnull().sum())
datasets={
    "Diabetes": df_d,
    "Social": df_s,
    "Titanic": df_t
```

```
}
for name, df in datasets.items():
    for col in df.columns:
        if df[col].dtype == 'object':
            df[col] = LabelEncoder().fit_transform(df[col])
    X = df.iloc[:, :-1]
    y = df.iloc[:, -1]
    X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)
    model = GaussianNB()
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    print(name, accuracy_score(y_test, y_pred))
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_8_1.py"
Diabetes
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1
Missing values
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Social Network Ads
    User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510    Male   19            19000          0
1  15810944    Male   35            20000          0
2  15668575  Female   26            43000          0
3  15603246  Female   27            57000          0
4  15804002    Male   19            76000          0
Missing values
User ID            0
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64

Titanic
   PassengerId  Survived  Pclass                                               Name     Sex   Age  SibSp  Parch            Ticket     Fare Cabin Embarked
0            1         0       3                            Braund, Mr. Owen Harris    male  22.0      1      0         A/5 21171   7.2500   NaN        S
1            2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1      0          PC 17599  71.2833   C85        C
2            3         1       3                             Heikkinen, Miss. Laina  female  26.0      0      0  STON/O2. 3101282   7.9250   NaN        S
3            4         1       1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1      0            113803  53.1000  C123        S
4            5         0       3                           Allen, Mr. William Henry    male  35.0      0      0            373450   8.0500   NaN        S
Missing values
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
Missing values (after analysis)
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
Diabetes 0.7662337662337663
Social 0.9125
Titanic 0.5251396648044693
```

PROBLEM 2: Apply the GAUSSIAN NAIVE BAYES CLASSIFIER on all three given datasets and analyze their performance. (HARDCODE)

CODE:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score
class NB_HC:
    def fit(self, X, y):
        self.classes = np.unique(y)
        self.mean = {}
        self.var = {}
        self.priors = {}
        for c in self.classes:
            X_c = X[y == c]
            self.mean[c] = np.mean(X_c, axis=0)
            self.var[c] = np.var(X_c, axis=0) + 1e-9
            self.priors[c] = X_c.shape[0] / X.shape[0]
    def predict(self, X):
        return np.array([self._predict(x) for x in X])
    def _predict(self, x):
        posteriors = []
        for c in self.classes:
            prior = np.log(self.priors[c])
            likelihood = np.sum(-0.5 * np.log(2 * np.pi *
self.var[c]) - ((x - self.mean[c]) ** 2) / (2 * self.var[c]))
            posteriors.append(prior + likelihood)
        return self.classes[np.argmax(posteriors)]

df_d = pd.read_csv('diabetes_dataset.csv')
df_d.fillna(df_d.mean(numeric_only=True), inplace=True)
df_s = pd.read_csv('Social_Network_Ads.csv')
df_s.fillna(df_s.mean(numeric_only=True), inplace=True)
df_t = pd.read_csv('titanic.csv')
df_t['Age'] = df_t['Age'].fillna(df_t['Age'].median())
df_t['Embarked'] =
df_t['Embarked'].fillna(df_t['Embarked'].mode()[0])
df_t = df_t.drop(columns=['Cabin'])
```

```
datasets = {
    "Diabetes": df_d,
    "Social": df_s,
    "Titanic": df_t
}
for name, df in datasets.items():
    for col in df.columns:
        if df[col].dtype == 'object':
            df[col] = LabelEncoder().fit_transform(df[col])
    X = df.iloc[:, :-1].values
    y = df.iloc[:, -1].values
    X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)

    model = NB_HC()
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    print(name, accuracy_score(y_test, y_pred))
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_8_3.py"
Diabetes
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1
Missing values
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Social Network Ads
    User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510    Male   19            19000          0
1  15810944    Male   35            20000          0
2  15668575  Female   26            43000          0
3  15603246  Female   27            57000          0
4  15804002    Male   19            76000          0
Missing values
User ID            0
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64

Titanic
   PassengerId  Survived  Pclass                                               Name     Sex   Age  SibSp  Parch            Ticket     Fare Cabin Embarked
0            1         0       3                            Braund, Mr. Owen Harris    male  22.0      1      0         A/5 21171   7.2500   NaN        S
1            2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1      0          PC 17599  71.2833   C85        C
2            3         1       3                             Heikkinen, Miss. Laina  female  26.0      0      0  STON/O2. 3101282   7.9250   NaN        S
3            4         1       1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1      0            113803  53.1000  C123        S
4            5         0       3                           Allen, Mr. William Henry    male  35.0      0      0            373450   8.0500   NaN        S
Missing values
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
Missing values (after analysis)
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
Diabetes 0.7662337662337663
Social 0.9125
Titanic 0.5251396648044693
```

PROBLEM 3: Apply the SVM on all three given datasets and analyze their performance.

CODE:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

df_d = pd.read_csv('diabetes_dataset.csv')
print("Diabetes")
print(df_d.head())
print("Missing values")
print(df_d.isnull().sum())
df_d.fillna(df_d.mean(numeric_only=True), inplace=True)

df_s = pd.read_csv('Social_Network_Ads.csv')
print("\nSocial Network Ads")
print(df_s.head())
print("Missing values")
print(df_s.isnull().sum())
df_s.fillna(df_s.mean(numeric_only=True), inplace=True)

df_t = pd.read_csv('titanic.csv')
print("\nTitanic")
print(df_t.head())
print("Missing values")
print(df_t.isnull().sum())
df_t['Age'] = df_t['Age'].fillna(df_t['Age'].median())
df_t['Embarked'] =
df_t['Embarked'].fillna(df_t['Embarked'].mode()[0])
df_t = df_t.drop(columns=['Cabin'])
print("Missing values (after analysis)")
print(df_t.isnull().sum())

datasets={
    "Diabetes": df_d,
    "Social": df_s,
    "Titanic": df_t
```

```
}

for name, df in datasets.items():
    for col in df.columns:
        if df[col].dtype == 'object':
            df[col] = LabelEncoder().fit_transform(df[col])
    X = df.iloc[:, :-1]
    y = df.iloc[:, -1]
    X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)
    model = SVC(kernel='rbf')
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    print(name, accuracy_score(y_test, y_pred))
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_8_2.py"
Diabetes
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1
Missing values
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Social Network Ads
    User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510    Male   19            19000          0
1  15810944    Male   35            20000          0
2  15668575  Female   26            43000          0
3  15603246  Female   27            57000          0
4  15804002    Male   19            76000          0
Missing values
User ID            0
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64


Titanic
   PassengerId  Survived  Pclass                                               Name     Sex   Age  SibSp  Parch            Ticket     Fare Cabin Embarked
0            1         0       3                            Braund, Mr. Owen Harris    male  22.0      1      0         A/5 21171   7.2500   NaN        S
1            2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1      0          PC 17599  71.2833   C85        C
2            3         1       3                             Heikkinen, Miss. Laina  female  26.0      0      0  STON/O2. 3101282   7.9250   NaN        S
3            4         1       1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1      0            113803  53.1000  C123        S
4            5         0       3                           Allen, Mr. William Henry    male  35.0      0      0            373450   8.0500   NaN        S
Missing values
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
Missing values (after analysis)
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
Diabetes 0.7337662337662337
Social 0.925
Titanic 0.6927374301675978
```

PROBLEM 4: Apply the SVM on all three given datasets and analyze their performance. (HARDCODE)

CODE:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score

class SVM_HC:
    def __init__(self, lr=0.001, lambda_param=0.01, n_iters=1000):
        self.lr = lr
        self.lambda_param = lambda_param
        self.n_iters = n_iters

    def fit(self, X, y):
        y_ = np.where(y <= 0, -1, 1)
        n_samples, n_features = X.shape
        self.w = np.zeros(n_features)
        self.b = 0

        for _ in range(self.n_iters):
            for idx, x_i in enumerate(X):
                condition = y_[idx] * (np.dot(x_i, self.w) - self.b) >= 1
                if condition:
                    self.w -= self.lr * (2 * self.lambda_param * self.w)
                else:
                    self.w -= self.lr * (2 * self.lambda_param * self.w - np.dot(x_i, y_[idx]))
                    self.b -= self.lr * y_[idx]

    def predict(self, X):
        linear_output = np.dot(X, self.w) - self.b
        return np.where(linear_output >= 0, 1, 0)

df_d = pd.read_csv('diabetes_dataset.csv')
print("Diabetes")
print(df_d.head())
print("Missing values")
print(df_d.isnull().sum())
df_d.fillna(df_d.mean(numeric_only=True), inplace=True)
```

```python
df_s = pd.read_csv('Social_Network_Ads.csv')
print("\nSocial Network Ads")
print(df_s.head())
print("Missing values")
print(df_s.isnull().sum())
df_s.fillna(df_s.mean(numeric_only=True), inplace=True)
df_t = pd.read_csv('titanic.csv')
print("\nTitanic")
print(df_t.head())
print("Missing values")
print(df_t.isnull().sum())
df_t['Age'] = df_t['Age'].fillna(df_t['Age'].median())
df_t['Embarked'] =
df_t['Embarked'].fillna(df_t['Embarked'].mode()[0])
df_t = df_t.drop(columns=['Cabin'])
print("Missing values (after analysis)")
print(df_t.isnull().sum())
datasets = {
    "Diabetes": df_d,
    "Social": df_s,
    "Titanic": df_t
}

for name, df in datasets.items():
    for col in df.columns:
        if df[col].dtype == 'object':
            df[col] = LabelEncoder().fit_transform(df[col])
    X = df.iloc[:, :-1].values
    y = df.iloc[:, -1].values
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)
    model = SVM_HC()
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    print(name, accuracy_score(y_test, y_pred))
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_8_4.py"
Diabetes
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1
Missing values
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Social Network Ads
    User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510    Male   19            19000          0
1  15810944    Male   35            20000          0
2  15668575  Female   26            43000          0
3  15603246  Female   27            57000          0
4  15804002    Male   19            76000          0
Missing values
User ID            0
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64

Titanic
   PassengerId  Survived  Pclass                                               Name     Sex   Age  SibSp  Parch            Ticket     Fare Cabin Embarked
0            1         0       3                            Braund, Mr. Owen Harris    male  22.0      1      0         A/5 21171   7.2500   NaN        S
1            2         1       1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1      0          PC 17599  71.2833   C85        C
2            3         1       3                             Heikkinen, Miss. Laina  female  26.0      0      0  STON/O2. 3101282   7.9250   NaN        S
3            4         1       1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1      0            113803  53.1000  C123        S
4            5         0       3                           Allen, Mr. William Henry    male  35.0      0      0            373450   8.0500   NaN        S
Missing values
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
Missing values (after analysis)
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
Diabetes 0.7662337662337663
Social 0.875
Titanic 0.09497206703910614
```